PEIZHONG JU

PARTICULARS

Current Position

Assistant Professor — Lexington, KY, USA
Department of Computer Science at **University of Kentucky** — *2024 - present*

Previous Positions

Postdoctoral Scholar — Columbus, OH, USA
ECE department at **The Ohio State University** — *2021 - 2024*
NSF AI Institute on Future Edge Networks and Distributed Intelligence (**AI-EDGE**)

Education

Purdue University — West Lafayette, IN, USA
Ph. D. in Electrical and Computer Engineering — *2016 - 2021*

Peking University — Beijing, China
B. Sc. in Electrical Engineering — *2012 - 2016*

Research Interests

My research interests span the areas of machine learning, smart grid, and wireless communication. My current research topics are about analyzing the performance of various machine learning models (such as generative AI) and designing new/better machine learning algorithms.

ACADEMIC HONORS

- Best Paper Award, 13th ACM International Conference on Future Energy Systems (ACM e-Energy 2022)
- The Bilsland Dissertation Fellowship, Purdue University, 2021
- Best Paper Award Finalist of 9th ACM International Conference on Future Energy Systems (ACM e-Energy 2018)
- Outstanding Graduate Award of EE department of Peking University, 2016 (the only one that year)
- Outstanding Graduate Award of Peking University, 2016
- IEEE Student Grant, 2015
- Research Excellence Award, Peking University, 2015
- China National Scholarship, 2015
- Samsung Scholarship, 2014
- EMC Scholarship, 2013

RESEARCH EXPERIENCE

- **Postdoctoral Scholar, the Ohio State University**, supervisor: Prof. Ness B. Shroff.
  - Multi-agent Reinforcement Learning, Multi-task supervised learning, Generalization Performance Analysis. (Oct. 2021 - Aug. 2024)
- **Research Assistant, Purdue University**, advisor: Prof. Xiaojun Lin.
  - Overfitting in Machine Learning (Oct. 2019 - Sept. 2021)
  - Market Design for Renewable in Power Distribution Level (Aug. 2018 - Oct. 2019)
  - Voltage Control in Power Grid with Renewable (Aug. 2016 - Aug. 2018)
- **Undergraduate Research Assistant, Peking University, China**, advisor: Prof. Xiang Cheng.
  - Antenna Grouping for Massive MIMO (Sep. 2015 - June 2016)
  - Self-Interference Cancellation for Full-Duplex (Oct. 2013 - Sep. 2015)

PUBLICATIONS

Peer-Reviewed Conference Papers

1. Fairoz Nower Khan, Nabuat Zaman Nahim, Ruiquan Huang, Haibo Yang, and **Peizhong Ju**, "Flow Matching for Offline Reinforcement Learning with Discrete Actions," 40th *Conference on Neural Information Processing Systems (NeurIPS'26)*, 2026 (accepted).
2. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, "Optimal Parallel Scheduling under Concave Speedup Functions," 27th *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'26)*, 2026 (accepted).
3. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, "Priority-Aware Encoding for Bandwidth-Efficient Real-Time Classification in 5G Networks," 24th *International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt'26)*.
4. Ashutosh Soni, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, "An LP-based Sampling Policy for Multi-Armed Bandits with Side-Observations and Stochastic Availability," 24th *International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt'26)*.
5. Moghis Fereidouni, Muhammad Umair Haider, **Peizhong Ju**, and A.B. Siddique, "Evaluating Sparse Autoencoders for Monosemantic Representation," Findings of 19th *Conference of the European Chapter of the Association for Computational Linguistics (EACL'26)*.
6. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, "Two Levels Are All You Need: Simplifying Data Compression for Timely Edge Classification," 26th *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'25)*.
7. Srijith Nair, Michael Lin, **Peizhong Ju**, Amirreza Talebi, Elizabeth Serena Bentley, and Jia Liu, "FSL-SAGE: Accelerating Federated Split Learning via Smashed Activation Gradient Estimation," 42nd *International Conference on Machine Learning (ICML'25)*.
8. Junze Deng, Qinhang Wu, **Peizhong Ju**, Sen Lin, Yingbin Liang, and Ness Shroff, "Unlocking the Power of Rehearsal in Continual Learning: A Theoretical Perspective," 42nd *International Conference on Machine Learning (ICML'25)*.
9. Yining Li, **Peizhong Ju**, and Ness B. Shroff, "How to Find the Exact Pareto Front for Multi-Objective MDPs?" 13th *International Conference on Learning Representations (ICLR'25)*. (**Spotlight Presentation**) (spotlight presentations rate: 5%))
10. Yuchen Liang, **Peizhong Ju**, Yingbin Liang, and Ness B. Shroff, "Theory on Score-Mismatched Diffusion Models and Zero-Shot Conditional Samplers," 13th *International Conference on Learning Representations (ICLR'25)*.
11. Yuchen Liang, **Peizhong Ju**, Yingbin Liang, and Ness B. Shroff, "Broadening Target Distributions for Accelerated Diffusion Models via a Novel Analysis Approach," 13th *International Conference on Learning Representations (ICLR'25)*.
12. Mingjing Xu, **Peizhong Ju**, Jia Liu, and Haibo Yang, "PSMGD: Periodic Stochastic Multi-Gradient Descent for Fast Multi-Objective Optimization," 39th *AAAI Conference on Artificial Intelligence (AAAI'25)*.
13. **Peizhong Ju**, Haibo Yang, Jia Liu, Yingbin Liang, and Ness B. Shroff, "Can We Theoretically Quantify the Impacts of Local Updates on the Generalization Performance of Federated Learning?" 25th *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'24)*.
14. Chengzhang Li, **Peizhong Ju**, Atilla Eryilmaz, and Ness B. Shroff, "Efficient Multi-dimensional Compression for Network-edge Classification," 25th *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'24)*.
15. **Peizhong Ju**, Arnob Ghosh, and Ness B. Shroff, "Achieving Fairness in Multi-Agent MDP Using Reinforcement Learning," 12th *International Conference on Learning Representations (ICLR'24)*.
16. Yining Li, **Peizhong Ju**, and Ness B. Shroff, "Achieving Sample and Computational Efficient Reinforcement Learning by Action Space Reduction via Grouping," 12th *International Conference on Learning Representations (ICLR'24)*.
17. Sen Lin, **Peizhong Ju (co-first author)**, Yingbin Liang, and Ness B. Shroff, "Theory on Forgetting and Generalization of Continual Learning," 40th *International Conference on Machine Learning (ICML'23)*.
18. **Peizhong Ju**, Yingbin Liang, and Ness B. Shroff, "Theoretical Characterization of the Generalization Performance of Overfitted Meta-Learning," 11th *International Conference on Learning Representations (ICLR'23)*.

19. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, "On the Generalization Power of Overfitted Three-Layer Neural Tangent Kernel Model," 36th *Conference on Neural Information Processing Systems (NeurIPS'22).*
20. **Peizhong Ju**, Xiaojun Lin, and Jianwei Huang, "Distribution-Level Markets under High Renewable Energy Penetration," 13th *ACM International Conference on Future Energy Systems (e-Energy'22)* (**Best Paper Award**).
21. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, "On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models," 38th *International Conference on Machine Learning (ICML'21).* (**Spotlight Presentation**)
22. **Peizhong Ju**, Xiaojun Lin, and Jia Liu, "Overfitting Can Be Harmless for Basis Pursuit, But Only to a Degree," 34th *Conference on Neural Information Processing Systems (NeurIPS'20).* (**Spotlight Presentation**) (acceptance rate: 20%, spotlight presentations: 3%).
23. **Peizhong Ju**, and Xiaojun Lin, "Adversarial Attacks to Distributed Voltage Control in Power Distribution Networks with DERs," *Proceedings of the* 9th *International Conference on Future Energy Systems (e-Energy'18).* (**Best Paper Award Finalist**).
24. **Peizhong Ju**, Meng Zhang, Xiang Cheng, and Liuqing Yang, "Generalized spatial modulation with transmit antenna grouping for massive MIMO," *Proceedings of 2017 IEEE International Conference on Communications (ICC'17).*
25. **Peizhong Ju**, Meng Zhang, Xiang Cheng, Cheng-Xiang Wang, and Liuqing Yang, "Generalized spatial modulation with transmit antenna grouping for correlated channels," *Proceedings of 2016 IEEE International Conference on Communications (ICC'16).*
26. **Peizhong Ju**, Miaowen Wen, Xiang Cheng, and Liuqing Yang, "An Effective Self-Interference Cancellation Scheme for Spatial Modulated Full Duplex Systems," *Proceedings of 2015 IEEE International Conference on Communications (ICC'15).*

Journal Papers

1. Nabuat Zaman Nahim, Fairoz Nower Khan, and **Peizhong Ju**, "A Survey of Flow Matching in Reinforcement Learning," *Transactions on Machine Learning Research (TMLR)*, 2026. **Survey Certification**.
2. Muhammad Umair Haider, Hammad Rizwan, Hassan Sajjad, **Peizhong Ju**, and A.B. Siddique, "Neurons Speak in Ranges: Breaking Free from Discrete Neuronal Attribution," *Transactions on Machine Learning Research (TMLR)*, 2026.
3. **Peizhong Ju**, Chengzhang Li (co-first author), Yingbin Liang, and Ness B. Shroff (the last two authors represent the entire AI-EDGE faculty team), "AI-EDGE: An NSF AI Institute for Future Edge Networks and Distributed Intelligence." *AI Magazine* 45.1 (2024): 29-34.
4. **Peizhong Ju**, Miaowen Wen, Xiang Cheng, and Liuqing Yang, "Achievable-Rate-Enhancing Self-Interference Cancellation for Full-Duplex Communications," *IEEE Transactions on Wireless Communications*, vol. 17 no. 12, pp. 8473-8484, 2018.
5. Weilin Qu, Meng Zhang, Xiang Cheng, and **Peizhong Ju**, "Generalized Spatial Modulation With Transmit Antenna Grouping for Massive MIMO," *IEEE Access*, vol. 5, pp. 26798-26807, 2017.

Pending Papers

1. Fairoz Nower Khan, Nabuat Zaman Nahim, and **Peizhong Ju**, "Flow Matching with Missing Data," *arXiv preprint arXiv:2607.28698 (2026).*
2. Amjad Mahdi Alqarni and **Peizhong Ju**, "FoA-SR: Faithful or Aesthetic? Profile-Aware Preference Optimization for Real-World Image Super-Resolution," *arXiv preprint arXiv:2606.10275 (2026).*
3. Fairoz Nower Khan, Nabuat Zaman Nahim, and **Peizhong Ju**, "Discrete Flow Matching for Offline-to-Online Reinforcement Learning," *arXiv preprint arXiv:2605.12379 (2026).*
4. Fairoz Nower Khan, Nabuat Zaman Nahim, Md Sajid Ahmed, Ruiquan Huang, and **Peizhong Ju**, "Discrete MeanFlow: One-Step Generation via Conditional Transition Kernels," *arXiv preprint arXiv:2605.12805 (2026).*
5. **Peizhong Ju**, Sen Lin, Mark S. Squillante, Yingbin Liang, and Ness B. Shroff, "Generalization Performance of Transfer Learning: Overparameterized and Underparameterized Regimes," *arXiv preprint arXiv:2306.04901 (2023).*
6. Ashutosh Soni, **Peizhong Ju**, Atilla Eryilmaz, Ness B. Shroff, "BeST – A Novel Source Selection Metric for Transfer Learning," *arXiv preprint arXiv:2501.10933 (2025)*

Book Chapters

1. **Peizhong Ju**, Xiaojun Lin, and Ness B. Shroff, "On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models," Chapter 3 of *Artificial Intelligence for Edge Computing*, Springer Nature, DOI: 10.1007/978-3-031-40787-1_3, August 2023.

Patents

1. Xiang Cheng, **Peizhong Ju**, and Miaowen Wen, "A method of self-interference cancellation scheme for full duplex systems," 2015100550190, State Intellectual Property Office of the People's Republic of China.

INVITED TALKS

1. "Flow Matching for Reinforcement Learning: From Continuous Flows to Discrete Decisions"
   Keeping Current (KCS) talks, Computer Science Department, University of Kentucky, Sep 2, 2026.
2. "Advancements in Multi-Objective Optimization: Exact Pareto Fronts in MDPs and Efficient Gradient Descent Methods"
   Keeping Current (KCS) talks, Computer Science Department, University of Kentucky, Feb 25, 2025.
3. "Machine Learning and Optimization in Multi-Agent Multi-Objective Frameworks"
   UK Applied Math Seminar, University of Kentucky, Oct 21, 2024.
4. "Overview of Reinforcement Learning", and "RL at the Edge: Fairness and Action Space Reduction"
   FutureG Summer Research Camp, Arizona State University, May 7, 2024.
5. "Achieving Fairness in Multi-Agent MDP Using Reinforcement Learning"
   AI-EDGE SPARKS Seminar, The Ohio State University, March 22, 2024.
6. "On the Generalization Power of the Overfitted Three-Layer Neural Tangent Kernel Model"
   The 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM-Statistics 2022), King's College London, 17-19 December 2022.
7. "Distribution Level Markets under High Renewable Energy Penetration"
   AIRS in the AIR Academic Series, Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), Shenzhen, China, August 23, 2022.
8. "On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models"
   Long Feng Science Forum, CUHK-Shenzhen, August 19, 2022.
9. "On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models"
   Joint Workshop between AI-EDGE and IBM, the Ohio State University, Columbus, June 15, 2022.
10. "On the Generalization Power of Overfitted Two-Layer Neural Tangent Kernel Models"
    TDAI deep learning summer school, the Ohio State University, Columbus, June 3, 2022.

MEDIA COVERAGE

- **The Ohio State University News** (Jul 20, 2023), "Future AI algorithms have potential to learn like humans, say researchers." A research by Sen Lin, Peizhong Ju, Yingbin Liang, and Ness Shroff on "continual learning" will be presented at the 40th International Conference on Machine Learning, Honolulu, Hawaii.

TEACHING & MENTORING EXPERIENCE

- **Instructor.** CS 463G: Introduction to Artificial Intelligence. Fall 2026, University of Kentucky. (34 undergraduate students)
- **Instructor.** CS 663: Artificial Intelligence. Fall 2026, University of Kentucky. (14 graduate students)
- **Instructor.** CS 465: Introduction to Generative AI. Spring 2026, University of Kentucky. (95 students)
- **Instructor.** CS 465: Introduction to Generative AI. Fall 2025, University of Kentucky. (40 students)
- **Instructor.** CS 463G: Introduction to Artificial Intelligence. Spring 2025, University of Kentucky. (45 students)
- **Instructor.** CS 660: Topics in AI: Generative Machine Learning. Fall 2024, University of Kentucky. (18 students)

- **Mentoring** several undergraduate students and one high school student in the summer research program held by AI-EDGE, 2024.
- **Secondary Instructor.** CSE-ECE 6461/6101: Introduction to Computer Communication Networks, Prof. Ness B. Shroff, Fall 2023, the Ohio State University.
- **Mentoring** a PhD student under the supervision of Prof. Ness B. Shroff and Prof. Atilla Eryilmaz at the Ohio State University (Dec. 2022 - present).
- **Mentoring** a female PhD student under the supervision of Prof. Ness B. Shroff at the Ohio State University (Oct. 2021 - present).
- **Mentoring** an undergraduate student and two high school students in the summer research program held by AI-EDGE, 2023.
- **Mentoring** an undergraduate student in Minority Serving Institutions (MSI) summer internships for AI-EDGE, 2022.
- **Teaching Assistant.** ECE 368: Data Structures, Prof. Cheng-Kok Koh, Spring 2020, Purdue University.
- **Teaching Assistant.** ECE 547: Introduction to Computer Communication Networks, Prof. Xiaojun Lin, Fall 2018 and Fall 2020, Purdue University.
- **Undergraduate Teaching Assistant.** Introduction to Computation, Prof. Yafei Dai, Fall 2012, Peking University, China.

SERVICE

- **Chair**
  - Session Chair, Session 3D: Learning for Wireless Systems, WiOpt 2026 (the 24th International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks), Columbus, Ohio, USA, Jun 5, 2026
  - EDAS Chair, WiOpt 2026 (the 24th International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks), Columbus, Ohio, USA, June 3-6, 2026
  - Session Chair, AAAI'25, Machine Learning 4, Room 119B at Pennsylvania Convention Center, Philadelphia, Pennsylvania, USA, February 28, 2025
- **Reviewer (Grants)**
  - National Science Foundation (NSF) Reviewer, 2025
- **Reviewer (Journals)**
  - Journal of Machine Learning Research (JMLR)
  - IEEE Transactions on Industrial Informatics (TII)
  - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
  - Array
  - Neurocomputing
  - Journal of Optimization Theory and Applications (JOTA)
  - IEEE Transactions on Information Theory (TIT)
  - IEEE Transactions on Signal Processing (TSP)
  - Transactions on Machine Learning Research (TMLR)
  - IEEE Transactions on Vehicular Technology (TVT)
  - IEEE Transactions on Wireless Communications (TWC)
  - IEEE Journal of Selected Topics in Signal Processing (J-STSP)
  - IEEE Transactions on Sustainable Computing (TSUSC)
  - IEEE Access
  - IEEE/ACM Transactions on Networking (TON)
  - Networks and Spatial Economics (NETS)
  - Internet Technology Letters
  - Transactions on Mobile Computing (TMC)
  - Frontiers of Information Technology & Electronic Engineering

  - ASME Journal of Dynamic Systems, Measurement and Control

- **Reviewer (Conferences)**
  - 43rd International Conference on Machine Learning (ICML'26), recognized as a **Silver Reviewer**
  - 2026 IEEE International Symposium on Information Theory (ISIT'26)
  - 29th International Conference on Artificial Intelligence and Statistics (AISTATS'26)
  - 14th International Conference on Learning Representations (ICLR'26)
  - 39th Conference on Neural Information Processing Systems (NeurIPS'25)
  - 18th ACM SIGGRAPH Conference and Exhibition on Computer Graphics and Interactive Techniques in Asia (SIGGRAPH Asia'25)
  - 42nd International Conference on Machine Learning (ICML'25)
  - 13th International Conference on Learning Representations (ICLR'25)
  - 28th International Conference on Artificial Intelligence and Statistics (AISTATS'25)
  - 38th Conference on Neural Information Processing Systems (NeurIPS'24)
  - 41st International Conference on Machine Learning (ICML'24)
  - 12th International Conference on Learning Representations (ICLR'24)
  - 37th Conference on Neural Information Processing Systems (NeurIPS'23)
  - 40th International Conference on Machine Learning (ICML'23)
  - 11th International Conference on Learning Representations (ICLR'23)
  - 36th Conference on Neural Information Processing Systems (NeurIPS'22)
  - 39th International Conference on Machine Learning (ICML'22)
  - 10th International Conference on Learning Representations (ICLR'22)
  - 35th Conference on Neural Information Processing Systems (NeurIPS'21)
  - 2021 National Conference on Communications (NCC'21)
  - 38th International Conference on Machine Learning (ICML'21)
  - IEEE Vehicular Technology Conference (VTC2015-Spring)
  - IEEE International Conference on Communications (ICC'16)

- **Program Committee**
  - 41st Annual AAAI Conference on Artificial Intelligence (AAAI'27)
  - 27th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'26)
  - IEEE International Conference on Computer Communications (INFOCOM) 2027, advisory TPC member
  - The IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) 2026 workshop: mHealth Security: Protecting Mobile and Connected Health Ecosystems (mHealthSec2026)
  - 40th Annual AAAI Conference on Artificial Intelligence (AAAI'26)
  - 26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc'25)
  - 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25)

- **Editor**
  - Guest Editor, Special Issue "Machine Learning in Networking Systems and Applications," *Electronics*, 2025

LANGUAGES

Proficient in English and Mandarin Chinese.